

Titanic Dataset Analysis Report

1. Introduction

This report analyzes the **Titanic dataset** to understand the factors influencing passenger survival. The main objectives are:

- Explore and clean the data for analysis.
 - Examine the relationship between survival and passenger features (e.g., age, sex, class, cabin, fare).
 - Visualize patterns and insights.
 - Summarize findings with actionable conclusions.
-

2. Data Description

The dataset contains information about passengers aboard the Titanic, including:

- PassengerId – Unique ID for each passenger
 - Survived – Survival status (0 = No, 1 = Yes)
 - Pclass – Ticket class (1, 2, 3)
 - Name, Sex, Age – Demographic information
 - SibSp, Parch – Family relations aboard
 - Ticket, Fare – Ticket details
 - Cabin – Cabin number
 - Embarked – Port of boarding
-

3. Data Cleaning & Preparation

The dataset contained some missing or inconsistent values. The cleaning steps included:

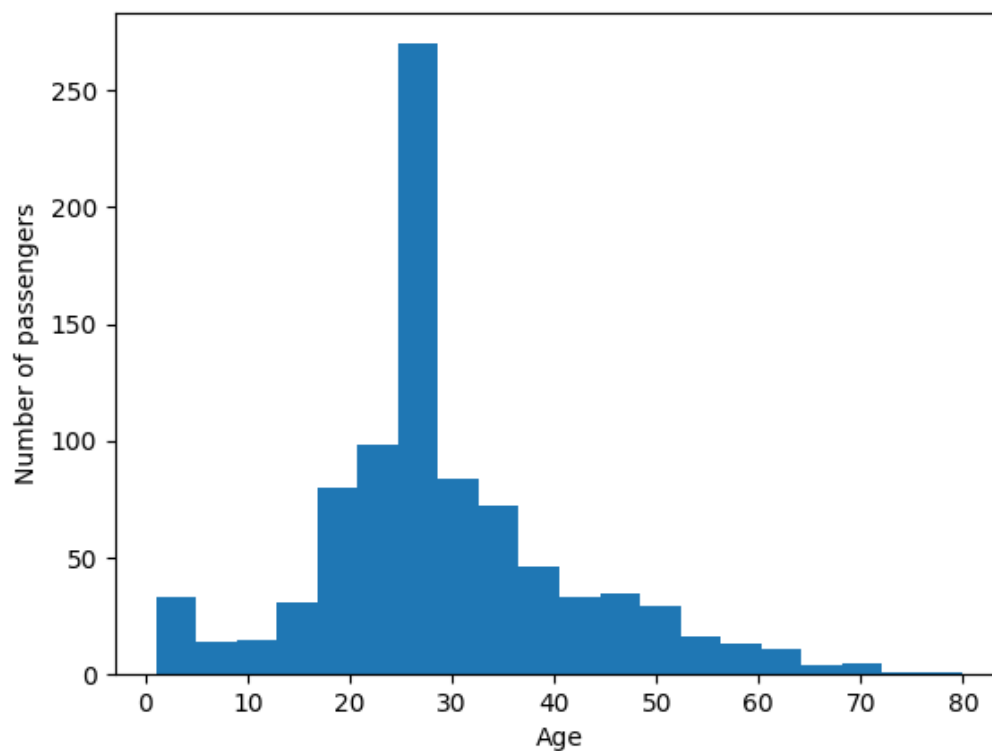
1. **Age:** Missing values replaced with median age; zeros replaced with median. Converted to integer.

2. **Cabin:** Missing values labeled as "Missing". Extracted first letter of cabin (Cabin_letter) for analysis.
3. **Embarked:** Missing values replaced with "Missing".

These steps ensure the dataset is ready for meaningful analysis.

4. Exploratory Data Analysis (EDA)

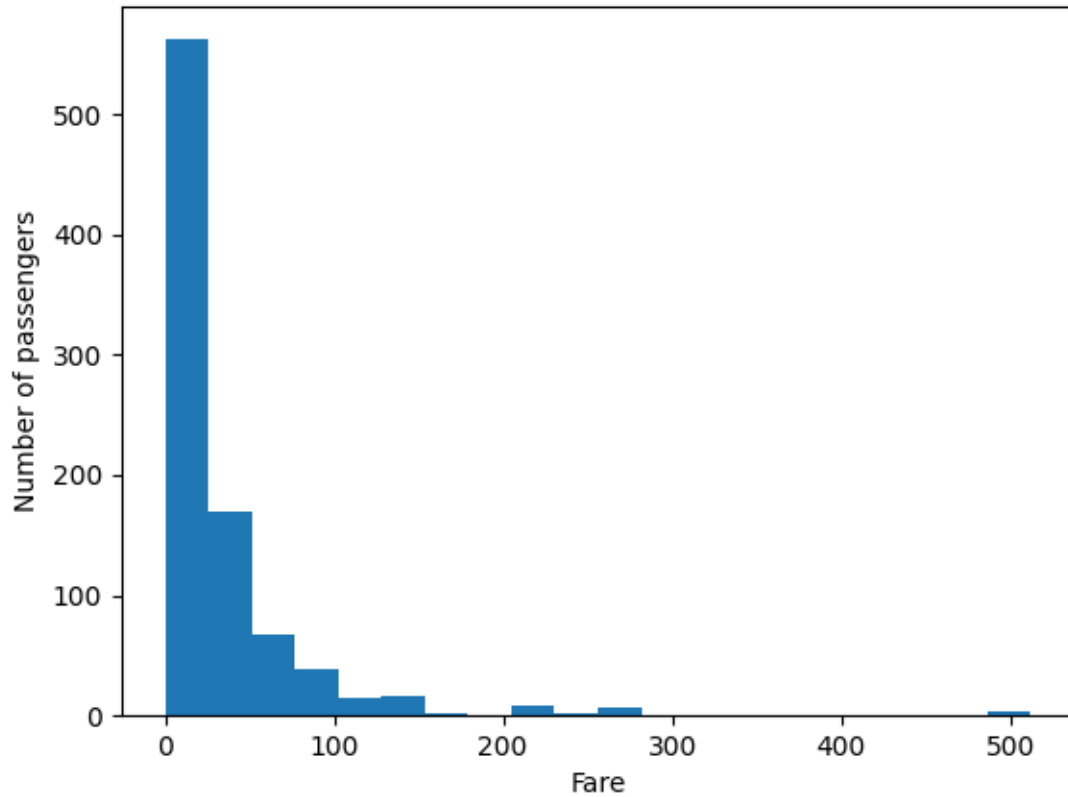
4.1 Age Distribution



Most passengers were adults (20–40 years).

Few children and seniors were onboard.

4.2 Fare Distribution



Majority of passengers paid lower fares.

A small number paid high fares, mostly first-class passengers.

4.3 Pclass Distribution

Third-class had the most passengers, followed by first and second class.

4.4 Sex Distribution

More male passengers than female passengers were onboard.

4.5 Cabin Distribution

A large number of passengers had missing cabin information, mostly third-class.

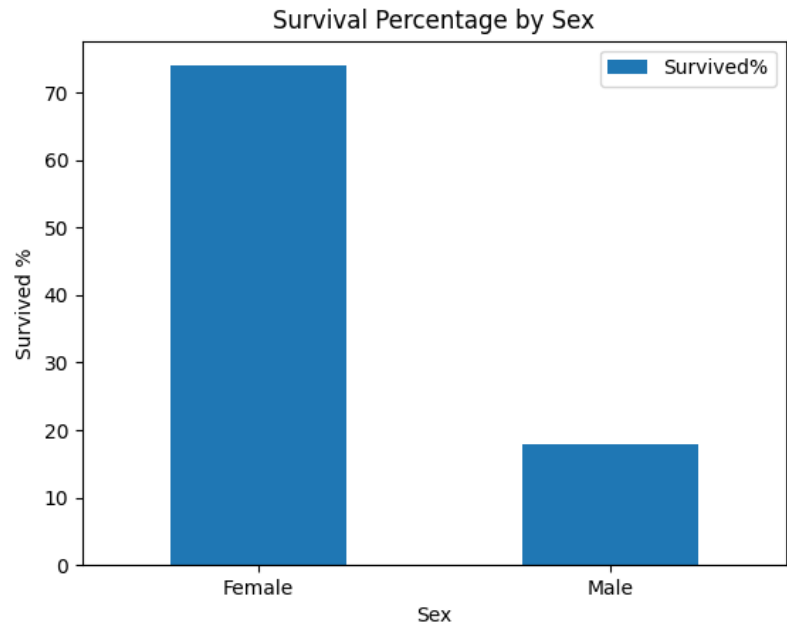
4.6 Survival Distribution

Survival rate overall was below 50%, indicating high fatality.

5. Analysis of Survival Factors

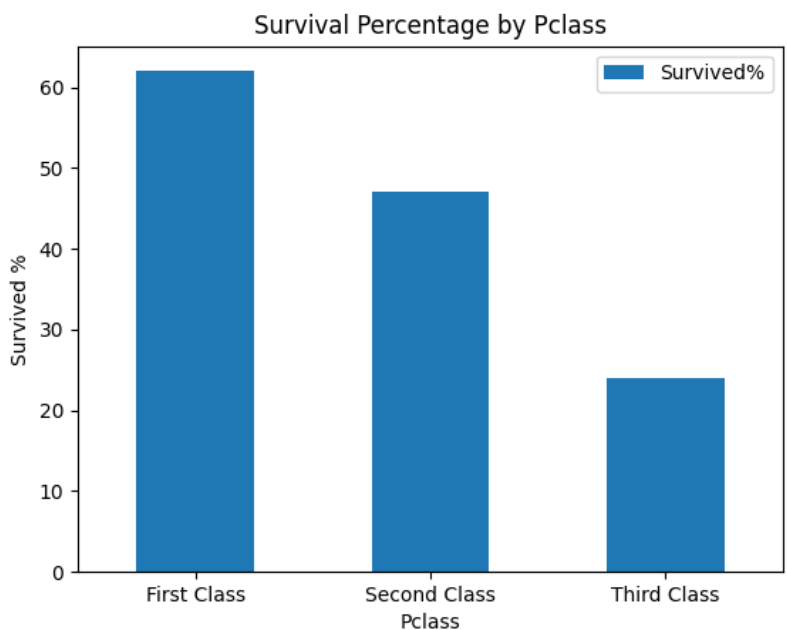
5.1 Survival by Sex

- **Findings:** Females had a significantly higher survival rate than males.
- **Interpretation:** Gender played a key role in survival due to the "women and children first" protocol.
- **Visualization:** Bar chart of survival % by sex.



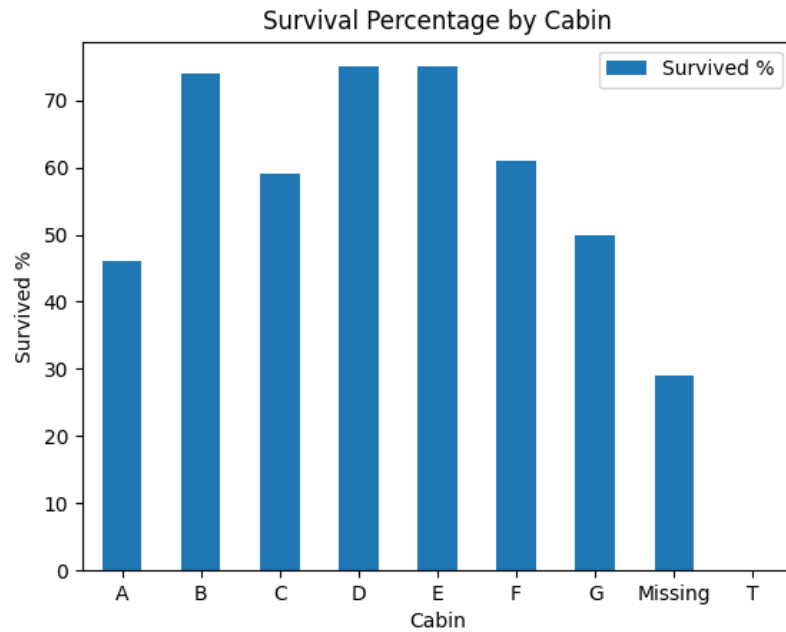
5.2 Survival by Pclass

- **Findings:** First-class passengers had higher survival rates than second and third class.
- **Interpretation:** Higher-class passengers had better access to lifeboats.

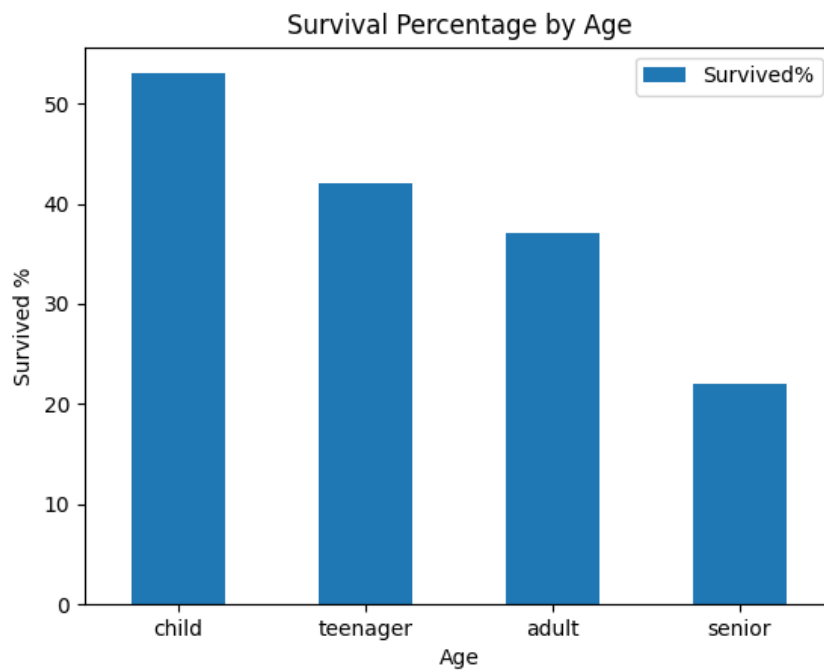


5.3 Survival by Cabin

- **Findings:** Cabins A–D had higher survival; missing cabins mostly correspond to lower classes.

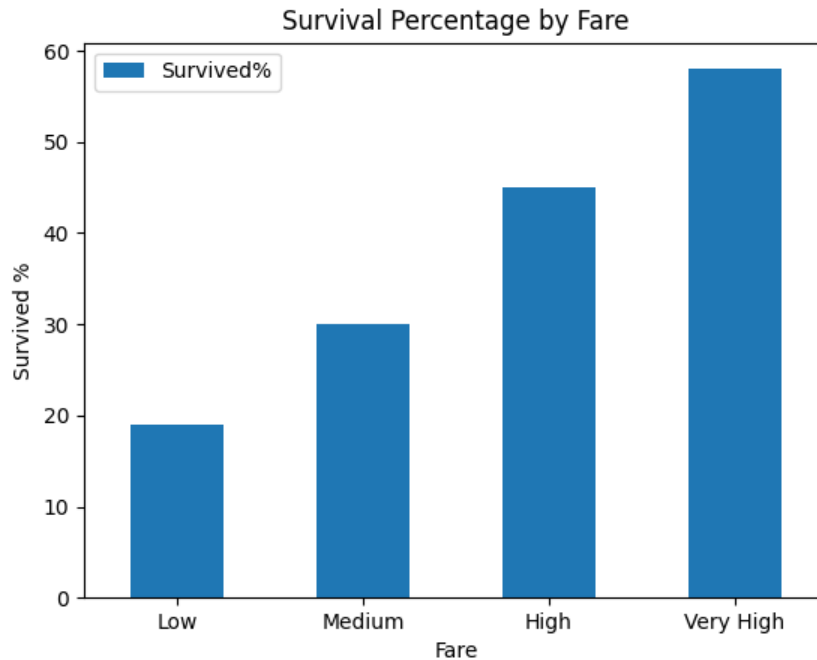


5.4 Survival by Age Group



- **Findings:** Children had slightly higher survival than adults. Seniors had lower survival.

5.5 Survival by Fare



- **Findings:** Higher fare passengers had better survival chances.
 - **Interpretation:** Fare correlates with class and access to lifeboats.
-

6. Conclusions

- **Sex:** Females survived more often than males.
- **Class:** First-class passengers had higher survival; third-class suffered most.
- **Age:** Children had slightly higher survival; seniors had lower survival.
- **Fare:** Higher fares improved survival chances.
- **Cabin Info:** Missing cabins were mostly third-class, suggesting limited cabin allocation affected survival.

Overall Insight: Survival on the Titanic was heavily influenced by social factors (sex, class, fare) rather than purely by chance.